

Automatic Number Plate Recognition System using YOLOv8 and Tesseract OCR

Overview

This project is an Automatic Number Plate Recognition (ANPR) system developed using Python. It detects vehicle number plates from images using YOLOv8 and extracts the plate number using Tesseract OCR. The system provides a graphical user interface (GUI) built with CustomTkinter and maintains a history of detections in a CSV database.

Features

- Upload vehicle images
 - Number plate detection using YOLOv8
 - Character recognition using Tesseract OCR
 - Image preprocessing and post-processing
 - Detection history with date and time
 - Automatic CSV database storage
 - Vehicle image saving
 - Output folder viewer
 - User-friendly GUI with image preview
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Technologies Used

- Python 3.12
 - YOLOv8 (Ultralytics)
 - OpenCV
 - Tesseract OCR
 - Pandas
 - CustomTkinter
 - Pillow (PIL)
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Project Structure

NumberPlateRecognition/

|

├── gui.py

├── detector.py

```
├── database.py
├── history.py
├── vehicle_database.csv
|
├── models/
|   └── best.pt
|
├── output/
|   ├── cropped_plate.jpg
|   ├── preprocessed_plate.jpg
|   └── detected_vehicle_*.jpg
|
└── README.md
```

⚙️ Workflow

Image Upload



YOLOv8 Plate Detection



Crop Number Plate



Image Preprocessing



Tesseract OCR



Post Processing



Detected Plate Number



CSV Database Storage



How to Run

1. Activate virtual environment:

```
venv\Scripts\activate
```

2. Install dependencies:

```
pip install ultralytics opencv-python pytesseract pandas pillow customtkinter
```

3. Run the application:

```
python gui.py
```

Sample Output

Detected Plate:

TG257602

Detection history is automatically stored in:

vehicle_database.csv

Future Enhancements

- Real-time camera support
 - Cloud database integration
 - Multi-vehicle detection
 - Advanced OCR models
 - Vehicle tracking system
 - Parking management integration
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